

KENYA HIGH 2025 PREDICTION



KCSE TEST EXAM

121/1

MATHEMATICS

PAPER 1

TIME: 2½ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

Instructions to candidates.

- Write your name, index number, admission number and name of your school in the spaces provided above
- Sign and write the date of the examination in the spaces provided above
- This paper consists of two sections: Section I and Section II.
- Answer all the questions in Section I and only five questions from Section II
- Show all the steps in your calculations, giving your answer at each stage in the spaces provided below each question
- Marks may be given for correct working even if the answer is wrong.
- Non-programmable silent electronic calculators and KNEC Mathematical tables may be used, except where stated otherwise.

For Examiner's use only.

Section I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Total

Section II

17	18	19	20	21	22	23	24	Total

Grand
Total

SECTION 1 (50 MARKS)

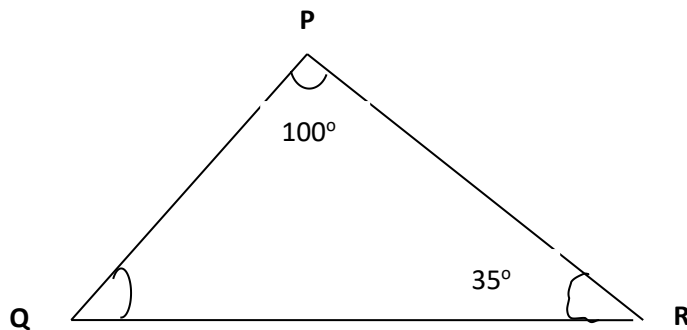
Answer all questions in the spaces provided.

1. A sum of money is divided between three men x, y and z in the ratio 5:3:1. If y has Shs. 700/= more than z, calculate how much x has. **(3 marks)**

2. Simplify the expression $\frac{12x^2 + ax - 6ax^2}{9x^2 - 4a^2}$ **(3 marks)**

3. Find the integral values of x of which **(3 marks)**
- $5 \leq 3x + 2$
- $3x - 14 \leq -2$

4. The figure below shows triangle PQR in which $PQ = 7\text{cm}$, angle $QPR = 100^\circ$ and angle $PRQ = 35^\circ$. Calculate to 2 decimal places the length of PR hence the area of triangle PQR.

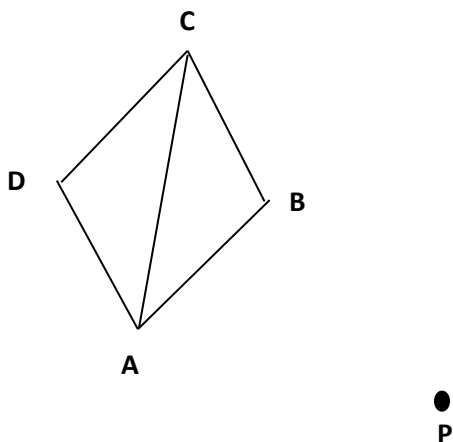


5. Evaluate without using a calculator

(3 marks)

$$\frac{23.4 - 2(5.2 + 5.3)}{3.2 \times 1.2}$$

6. The figure below represents a kite ABCD. Use a pair of compasses and ruler only to rotate it through -60° about point P. **(2 marks)**



7. Find the inverse of the matrix $\begin{pmatrix} 1 & 1 \\ 3 & 1 \end{pmatrix}$ hence determine the point of intersection of the lines.

$$y + x = 7$$

$$3x + y = 15$$

(4 marks)

8. A trader had a bag of rice, when he packed the rice in 6kg packets, he had 1 kg left over, when he packed the rice in 8kg packets, again he had 1kg left over. When he packed the rice in 9kg packets, he had 1kg left over. What is the smallest amount of rice that he must have had.

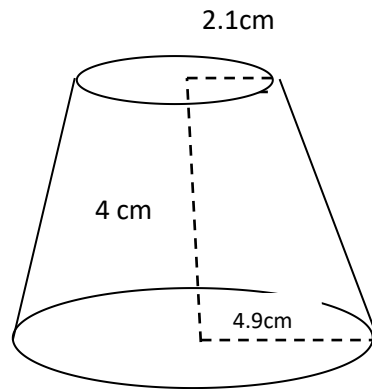
(3 marks)

9. A Histogram is drawn from the se of data given below. Complete the missing bars.(3 marks)

Marks	6 - 10	11 - 20	21 - 35	36 - 55	56 - 65
Frequency	8	14	18	24	10

10. Kinyua marked an article at Kshs. 1200 and sold it to a customer at a discount of 15%. Find the percentage profit he made if he had bought the article at Kshs. 900/=. (3 marks)

11. The diagram below represents a solid of a conical frustrum. (Use $\pi = \frac{22}{7}$)



Calculate the volume of the solid.

(4 marks)

12. Given that $a = \begin{pmatrix} -2 \\ 8 \end{pmatrix}$, $b = \begin{pmatrix} -6 \\ 4 \end{pmatrix}$ and $c = \begin{pmatrix} -4 \\ 2 \end{pmatrix}$ and given that $P = 4a - 8b + 6c$, find P /(3 marks)

13. Evaluate using tables of reciprocals

(3 marks)

$$\frac{5}{807} + \frac{1}{0.0591}$$

14. Solve for x in the following equation

(3 marks)

$$\left(\frac{1}{4} \right)^{x-2} = 2^{x+2}$$

15. Each interior angle of a regular polygon is four times greater than the exterior angle. How many sides does this polygon have? **(3 marks)**

16. A boat is at point P, a distance of 100km from the bottom of a hill. The angle of elevation of the top of the hill is 30° from P. The boat sails straight towards the hill to a point Q from where the angle of elevation to the top of the hill is now 60° . Calculate the distance PQ **(3 marks)**

SECTION II (50 MARKS)

Answer only five question from this section in the spaces provided.

17. Given that a line L_1 passes through the points A(-1, 5) and B (3, -1), find

a) The equation of line L_1 in the form $y = mx + c$ **(2 marks)**

b) The equation of a line L_2 , which is a perpendicular bisector of L_1 . Leave your answer in the form $ax + by = c$ where a, b, c are integers **(3 marks)**

c) Given that another line L_3 is parallel to L_2 and passes through point (-3, -5) and intersects lines L_1 at point P. Find the equation of L_3 in the form $ax + by + c = 0$ **(2 marks)**

d) The coordinates of the point of intersection of lines L_1 and L_3 **(3 marks)**

- 18.** Four towns P, R, T and S are such that R is 70km directly to the north of P and T is on a bearing of 280° from P at a distance of 75km. S is on a bearing of 320° from T and a distance of 45km.
- a) Using a scale of 1cm to represent 10km, make an accurate scale drawing to show the relative position of the towns. **(4 marks)**

- b) Using your diagram above, find the distance in km of **(3 marks)**
- (i) R from T
- (ii) S from R
- c) From your diagram, what is the compass bearing of **(3 marks)**
- (i) R from T

(ii) P from S

19. Nyaugenya bus leaves Port Victoria for Eldoret at 7:00 a.m. at an average speed of 80km/h. Climax bus leaves Eldoret towards Port Victoria at 7:30 a.m on the same day using the same route at an average speed of 60km/h. The distance from Eldoret to Port Victoria is 450km. After travelling for one hour and a half, climax bus developed a mechanical problem which took 45 minutes to repair before continuing at its speed in the same direction.

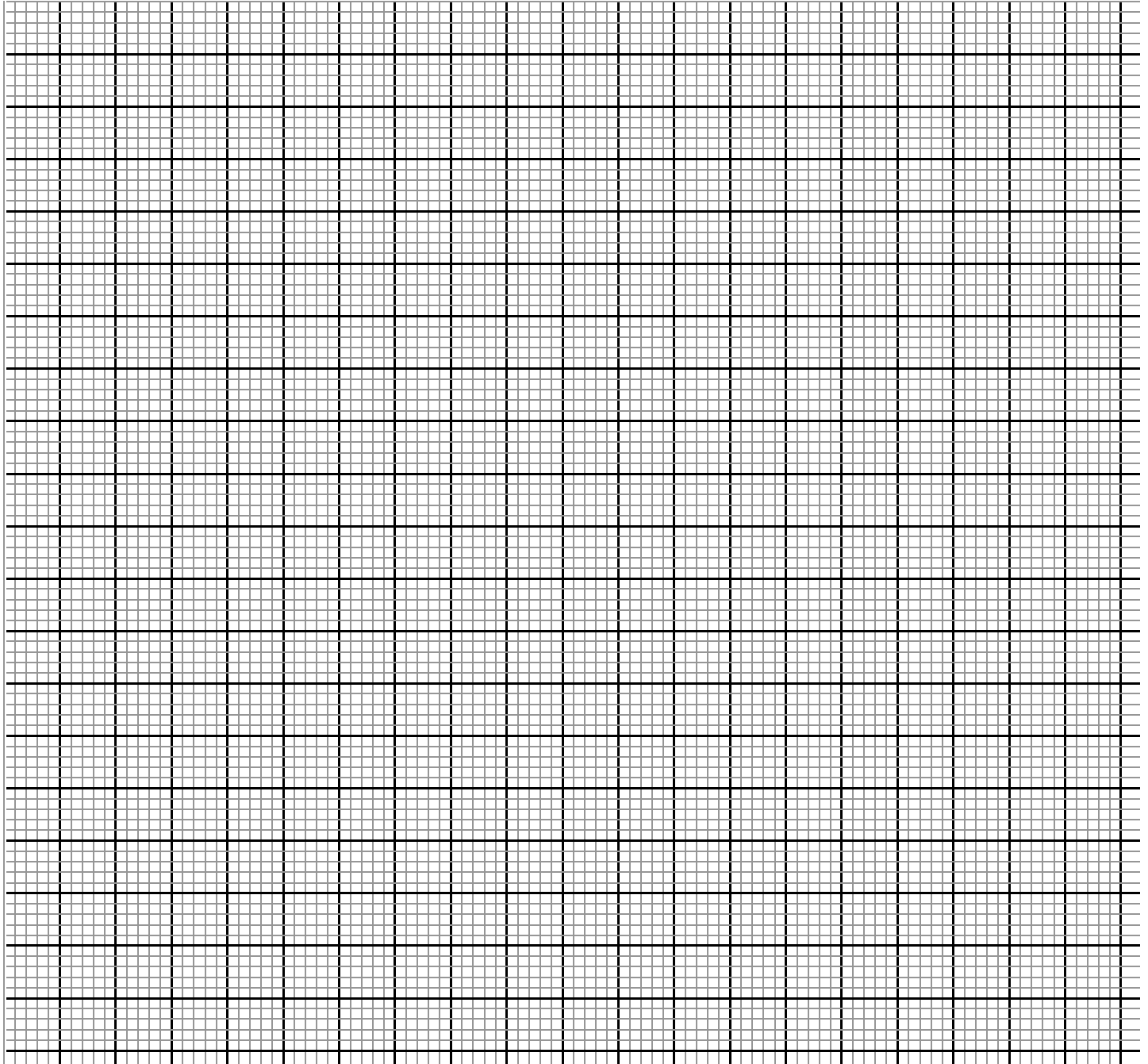
a) Determine the time when the two buses met. **(4 marks)**

b) Calculate the distance from Eldoret at the time of the two buses meeting. **(3 marks)**

c) For how long did the Nyaugenya bus stay in Eldoret before Climax bus arrived at Port Victoria.

(3 marks)

20. (a) On the grid provided draw triangle ABC with vertices A(0, -1), B(4, 3) and C(2, 2) (1 mark)



- b)** Draw triangle $A^1B^1C^1$, the image of ABC under a translation defined by the translation vector $T = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$.
Write down the coordinates $A^1B^1C^1$.
(3 marks)

- c)** $A^{11}B^{11}C^{11}$ is the image of $A^1B^1C^1$ under an enlargement, scale factor -2 , centre $(3, 1)$. On the same grid draw $A^{11}B^{11}C^{11}$ and write down its coordinates. **(4 marks)**

- d)** Draw $A^{111}B^{111}C^{111}$, the image of $A^{11}B^{11}C^{11}$ under reflection on the line $x = 0$ **(2 marks)**

21. The motion of a particle P moving along a straight line is described by the equation $S = (8 + 10t)t - t^3$

Calculate;

a) The distance when $t = 3$ sec. **(2 marks)**

b) The maximum velocity of the motion. **(4 marks)**

c) The acceleration of motion after 4 seconds. **(2 marks)**

d) The time at which the velocity is zero. **(2 marks)**

22. a) Use the trapezium rule with 8 ordinates to approximate the area under the curve $y = x^2 + x + 3$ and the x-axis between lines $x = -3$ and $x = 4$ (5 marks)

- b) Use the mid-ordinate rule with 5 strips to calculate the area under the curve $y = 3x^2 + 8$ and bounded by lines $y = 0$, $x = 1$ and $x = 6$. (5 marks)

23. Three people; A, B and C work together to make a certain number of tins. If person C was to work alone he will take $4\frac{4}{9}$ hours to complete the job. If all working together they will take 1 hour 40 min. to complete the job. They all started working together however person B left after the first 40 minutes, while person C left 20 minutes later. Person A took a further 1hr 46mins. Calculate how long it would take if all the tins were made by;

a) Person A alone

(6 marks)

b) Person B alone

(2 marks)

c) Person A and C alone

(2 marks)

24. In Busia County, a tailor bought a number of suits at a cost of Shs. 57,600/= from a wholesaler. Had he bought the same number of suits from a supermarket, it would have cost him Shs. 480/= less per unit. This would have enabled him to buy four extra suits for the same amount of money.

a) By letting the number of suits that the tailor bought to be n , write an expression in n for cost of a suit from;

(i) The wholesaler

(1 mark)

(ii) The supermarket

(1 mark)

b) Find the number of suits that the trailer bought

(4 marks)

c) The tailor later sold each suit for Shs. 720/= more than he paid for it. Determine the percentage profit he made.

(4 marks)